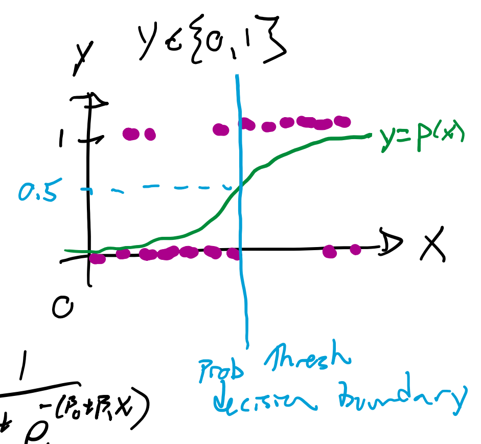
Wk4.2 Other classification methods

Thursday, October 1, 2026 9:19 AM

Classification: $Y = f(x) + \varepsilon$



3 approaches:

Full detail ① Directly model $p(x) = P(Y=1|X)$

↳ Logistic regression:

$$\underbrace{\ln\left(\frac{p}{1-p}\right)}_{\text{logit link}} = \beta_0 + \beta_1 X \rightarrow p(x) = \underbrace{\frac{1}{1+e^{-(\beta_0+\beta_1 X)}}}_{\text{logistic/sigmoid}}$$

Light detail ② Indirectly model $P(Y=1|X)$ via $\underset{\text{prob distr for } X}{P(X|Y=1)}$

by using $\boxed{\text{Baye's Rule}}$ ★

Generative models (X~sens)

↳ Discriminant analysis (DA), linear (LDA) & quadr (QDA)

classic stat history

↳ Naive Baye's, assumes $X = (X_1, X_2, \dots, X_p)$ has independent variables

③ Non-parametric

↳ KNNs & class dec trees/RF/GBM

ML accuracy

Also have GLM (generalized linear model)

Skip

general regr framework:

① LR: $\underset{\text{normal}}{Y} = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$

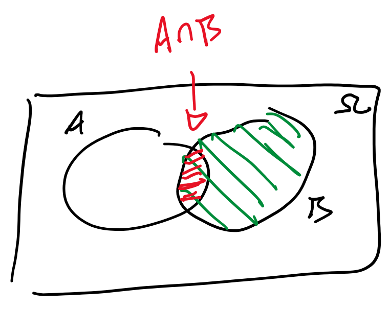
② LogR: $\underset{\text{binomial/Bernoulli}}{\text{logit}(p)} = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$

③ More.

Discriminant Analysis Model $P(Y=1|X)$ via $P(X|Y=1)$ & Baye's Rule.

Baye's Rule: $P(A|B) = \dfrac{P(B|A)P(A)}{P(B)}$

Solve $P(A|B)$



$$\boxed{\begin{aligned} P(A|B) &= \frac{P(A\cap B)}{P(B)} \\ P(B|A) &= \frac{P(A\cap B)}{P(A)} \end{aligned}}$$

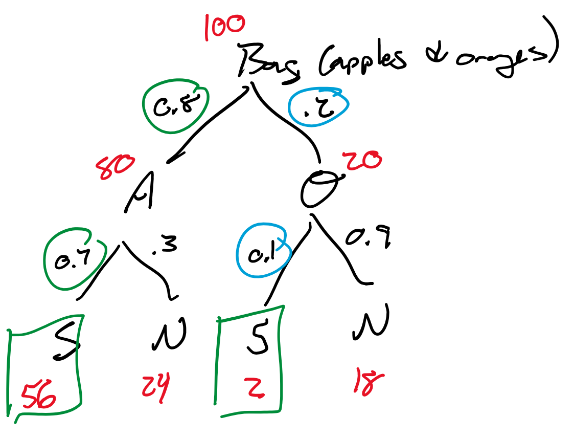
$P(A\cap B) = P(A|B)\,P(B)$
$= P(B|A)\,P(A)$

2 uses: ① Trade $P(A|B)$ for $P(B|A)$

② Info update:

$$\underset{\text{posterior (new info)}}{P(A|B)} = \frac{\overset{\text{likelihood}}{P(B|A)}\ \overset{\text{prior (have first)}}{P(A)}}{\underset{\text{Normal constant}}{P(B)}}$$

Ex Using Baye's Rule (Stat 145)



$$\underset{\text{prob of Apple draw given shiny}}{P(A|S)} = \frac{P(S|A)\,P(A)}{\underset{\text{total prob}}{P(S)}}$$

$$= \frac{0.7\,(0.8)}{P(S|A)P(A) + P(S|O)\,P(O)}$$

$$= \frac{0.7\,(0.8)}{0.7\,(0.8) + 0.1\,(0.2)}$$

$$= \frac{0.56}{0.56+0.02} = \boxed{\frac{56}{58}} = 0.966$$

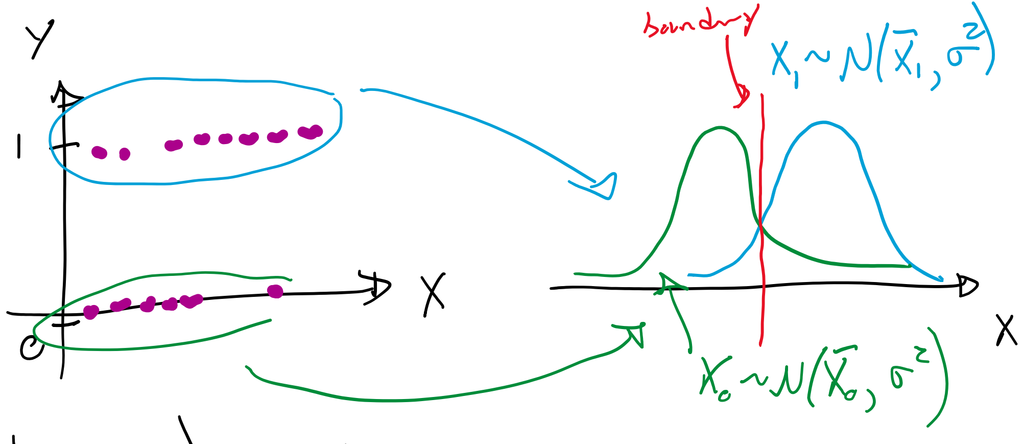
Discriminant Analysis

% of 1 classif

$$P(Y=1|X) \overset{BR}{=} \frac{P(X|Y=1)\,P(Y=1)}{P(X)}$$

$$= \frac{\boxed{P(X|Y=1)}\,P(Y=1)}{\boxed{P(X|Y=1)}\,P(Y=1) + \boxed{P(X|Y=0)}\,P(Y=0)}$$

Assumes $P(X|Y=1)$ is normally distributed
& $P(X|Y=0)$



2 ways: ① If assume $P(X|Y=i)$ have same variance, gives LDA

② If diff variance per $P(X|Y=i)$, gives QDA.

See text for full Stat & LA details.

Naive Bayes $X = (X_1, \dots, X_p)$, $p > 1$

Assume X_i independent.

$$P(Y=1|X) = \frac{\boxed{P(X|Y=1)}\,P(Y=1)}{\sum_{i=1}^{n} P(X|Y_i)}$$

X indep gives

$$P(X|Y=1) = \underbrace{P(X_1|Y=1)\,P(X_2|Y=1)\dots P(X_p|Y=1)}_{\text{p-separate normal distr.}}$$